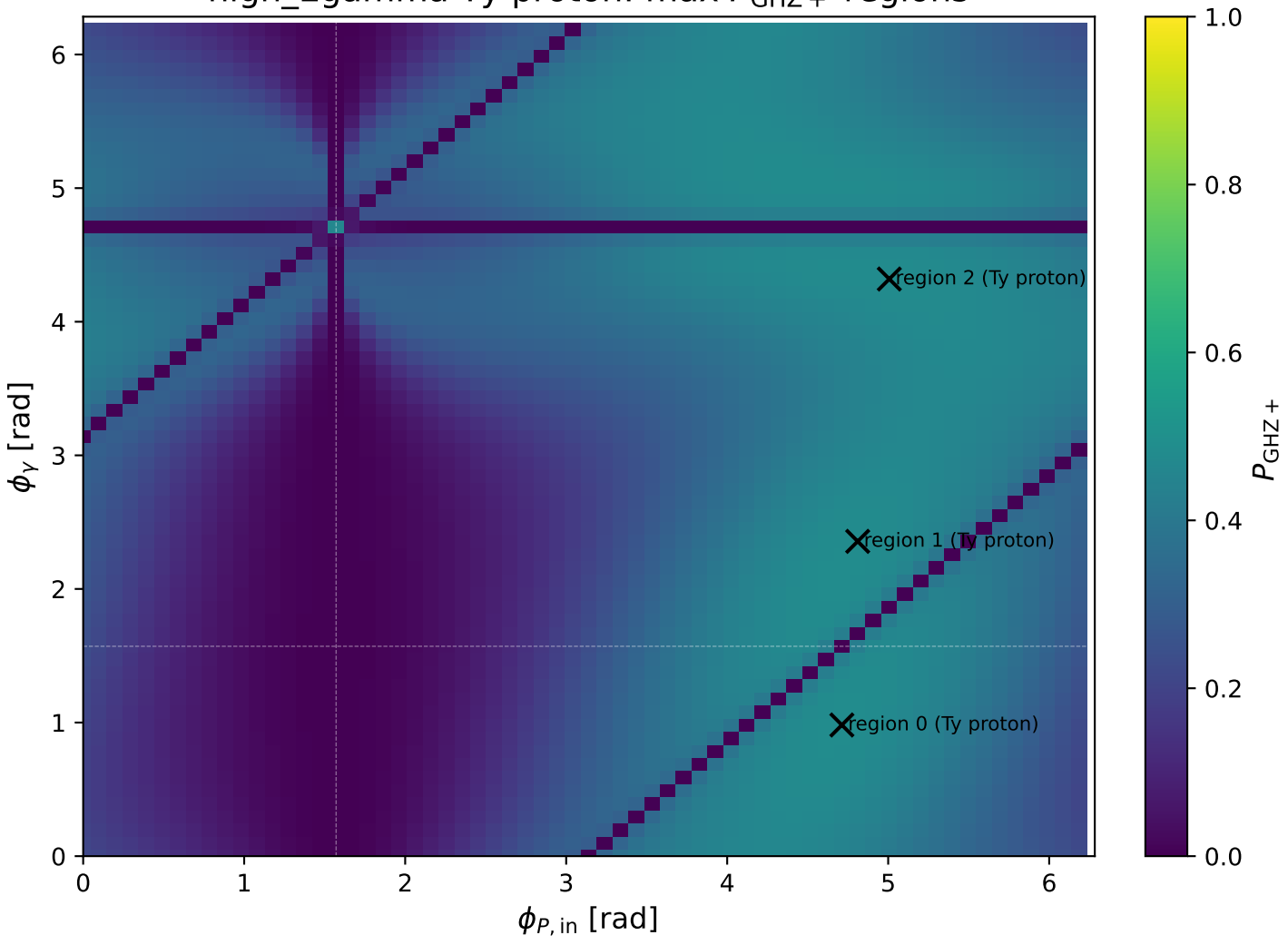
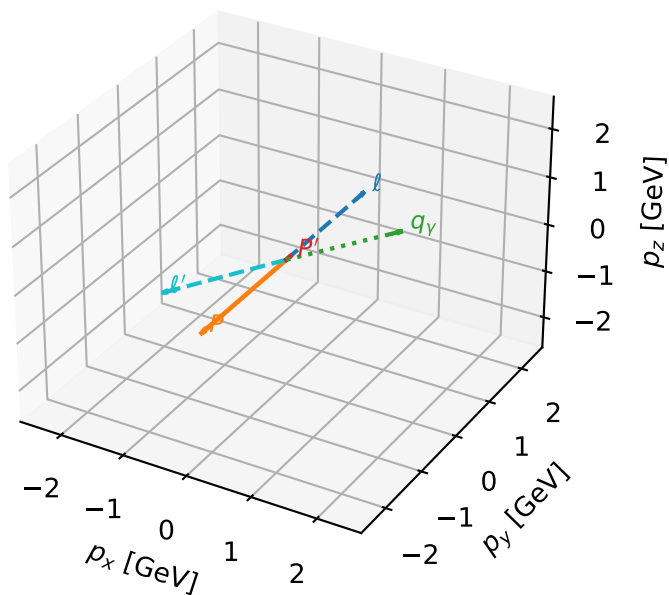


high_Egamma Ty proton: max $P_{\text{GHZ}+}$ regions



Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

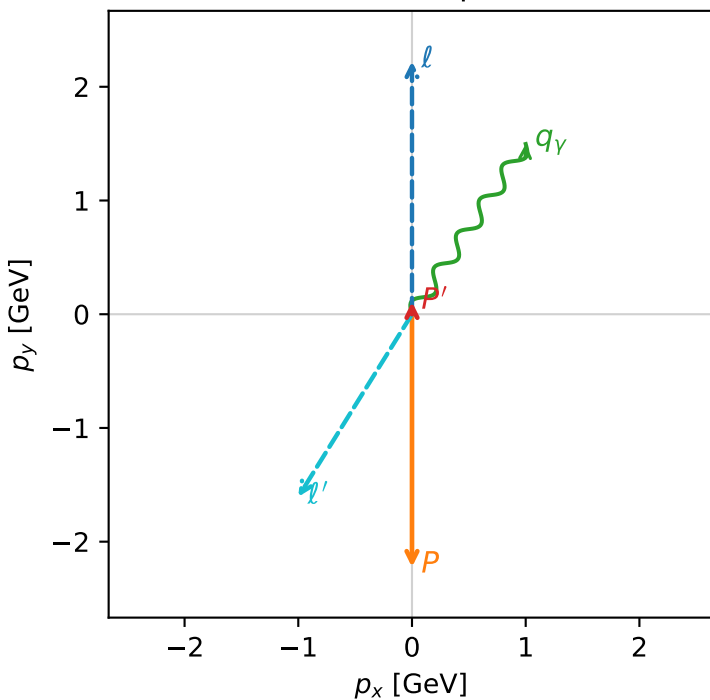


ghz_purity_high_egamma_ty_proton_region_0 $P_{\{\text{rm GHZ+}\}}=0.482446$
 region: high_Egamma_Ty_proton_region_0
 outgoing-state purity: 0.50004
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $\frac{1}{2}\sum_{h_\mu}\rho(h_\mu, T_{y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.108644$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.5708$, $\phi_p=4.71239$
 $E_\gamma=1.8$, $\phi_\gamma=0.981748$
 $Q^2=15.5469$, $x_B=0.834909$, $t=-3.28022$, $W^2=3.95402$, $y=0.902847$
 $\theta(\ell', \gamma)=178.171$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= 0.0000$ $p_y= 2.2231$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= -0.0000$ $p_y= -2.2231$ $p_z= 0.0000$
 ℓ' $E= 1.8942$ $p_x= -1.0000$ $p_y= -1.6053$ $p_z= 0.0000$
 P' $E= 0.9443$ $p_x= 0.0000$ $p_y= 0.1086$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= 1.0000$ $p_y= 1.4966$ $p_z= 0.0000$

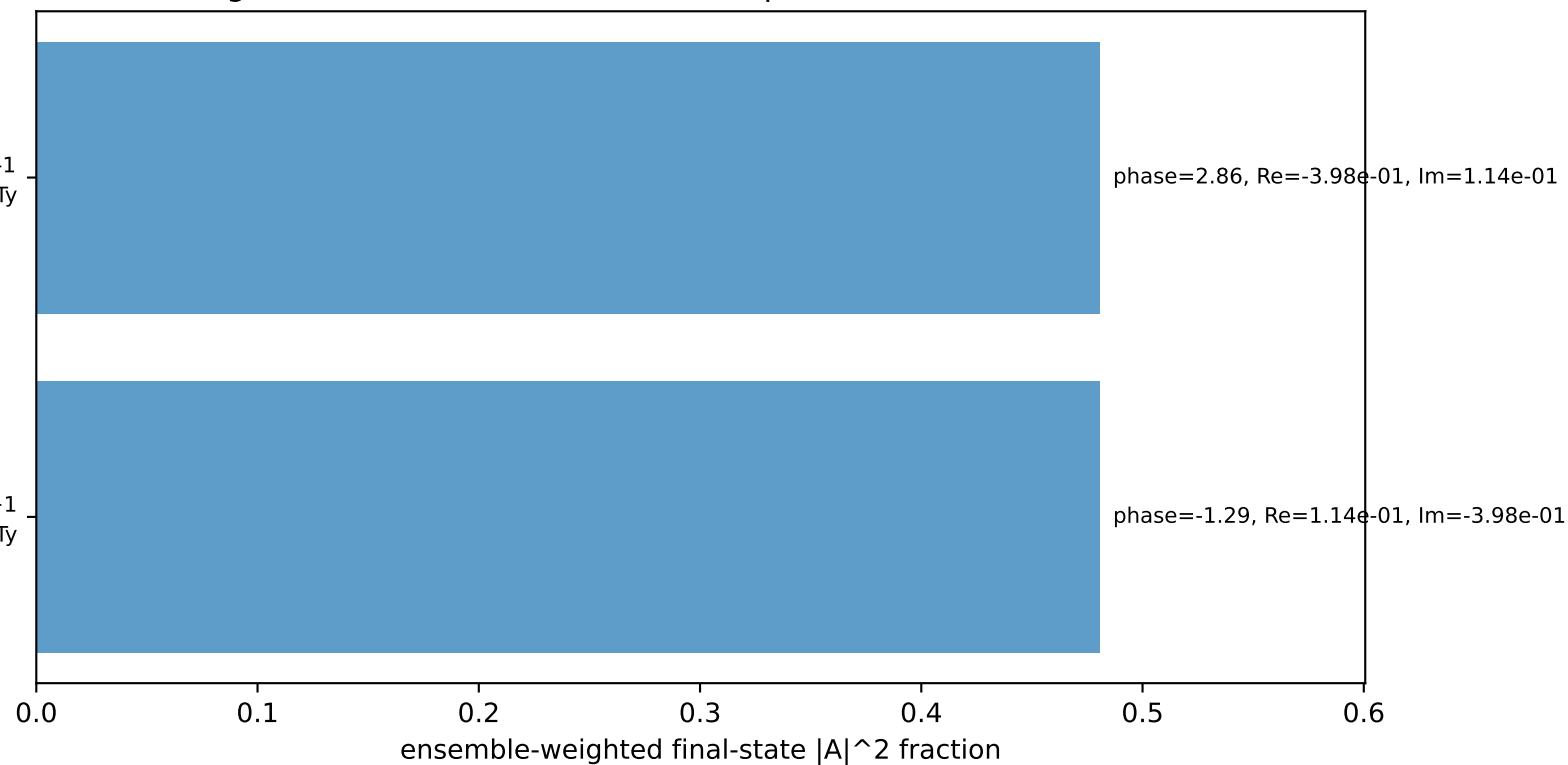
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=-1$
 muon $h=-1$, proton Ty

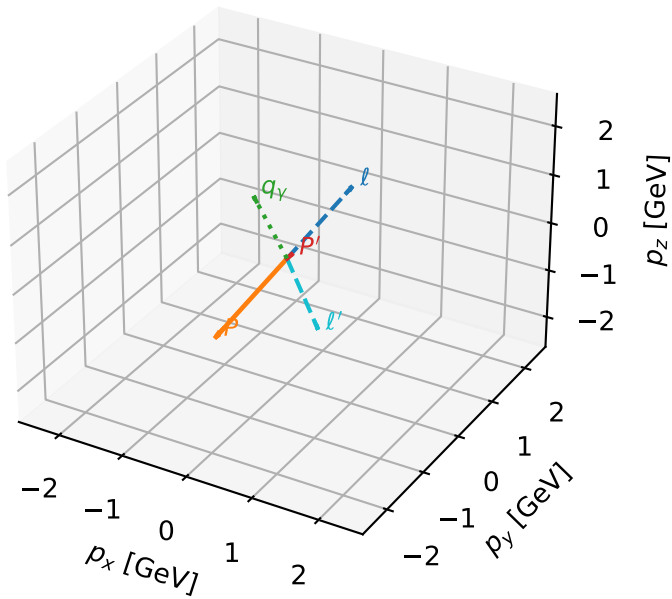
$h_e=+1, h_p=+1, h_\gamma=+1$
 muon $h=+1$, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=96.1%)



Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

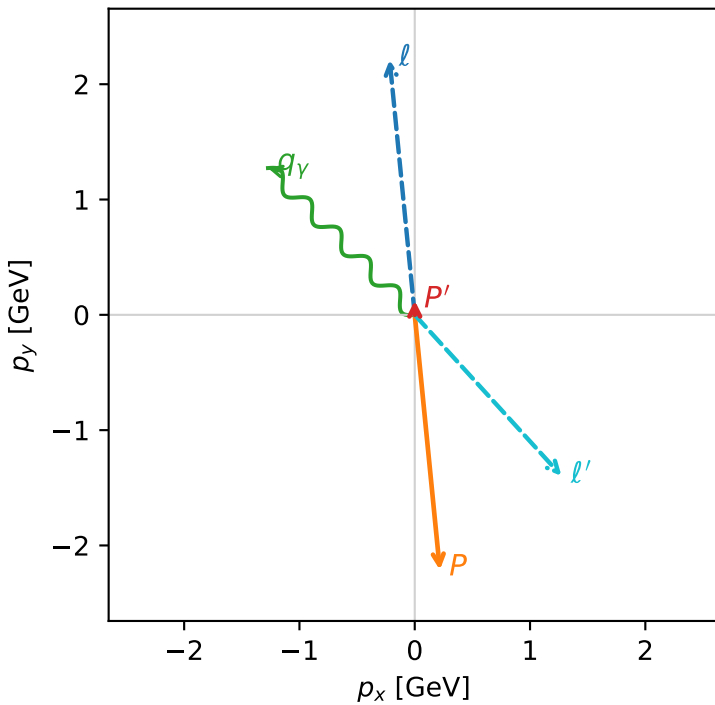


ghz_purity_high_egamma_ty_proton_region_1 $P_{\{\text{rm GHZ+}\}}=0.480795$
 region: high_Egamma_Ty_proton_region_1
 outgoing-state purity: 0.500036
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $\frac{1}{2} \sum_{h_\mu} \rho(h_\mu, T_{y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=0.123642$
 $\theta_{\text{in}}=1.57079$, $\phi_\ell=1.66897$, $\phi_P=4.81056$
 $E_\gamma=1.8$, $\phi_\gamma=2.35619$
 $Q^2=15.1349$, $x_B=0.830394$, $t=-3.35315$, $W^2=3.97111$, $y=0.883699$
 $\theta(\ell', \gamma)=177.348$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -0.2179$ $p_y= 2.2124$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.2179$ $p_y= -2.2124$ $p_z= 0.0000$
 ℓ' $E= 1.8924$ $p_x= 1.2728$ $p_y= -1.3964$ $p_z= 0.0000$
 P' $E= 0.9461$ $p_x= 0.0000$ $p_y= 0.1236$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -1.2728$ $p_y= 1.2728$ $p_z= 0.0000$

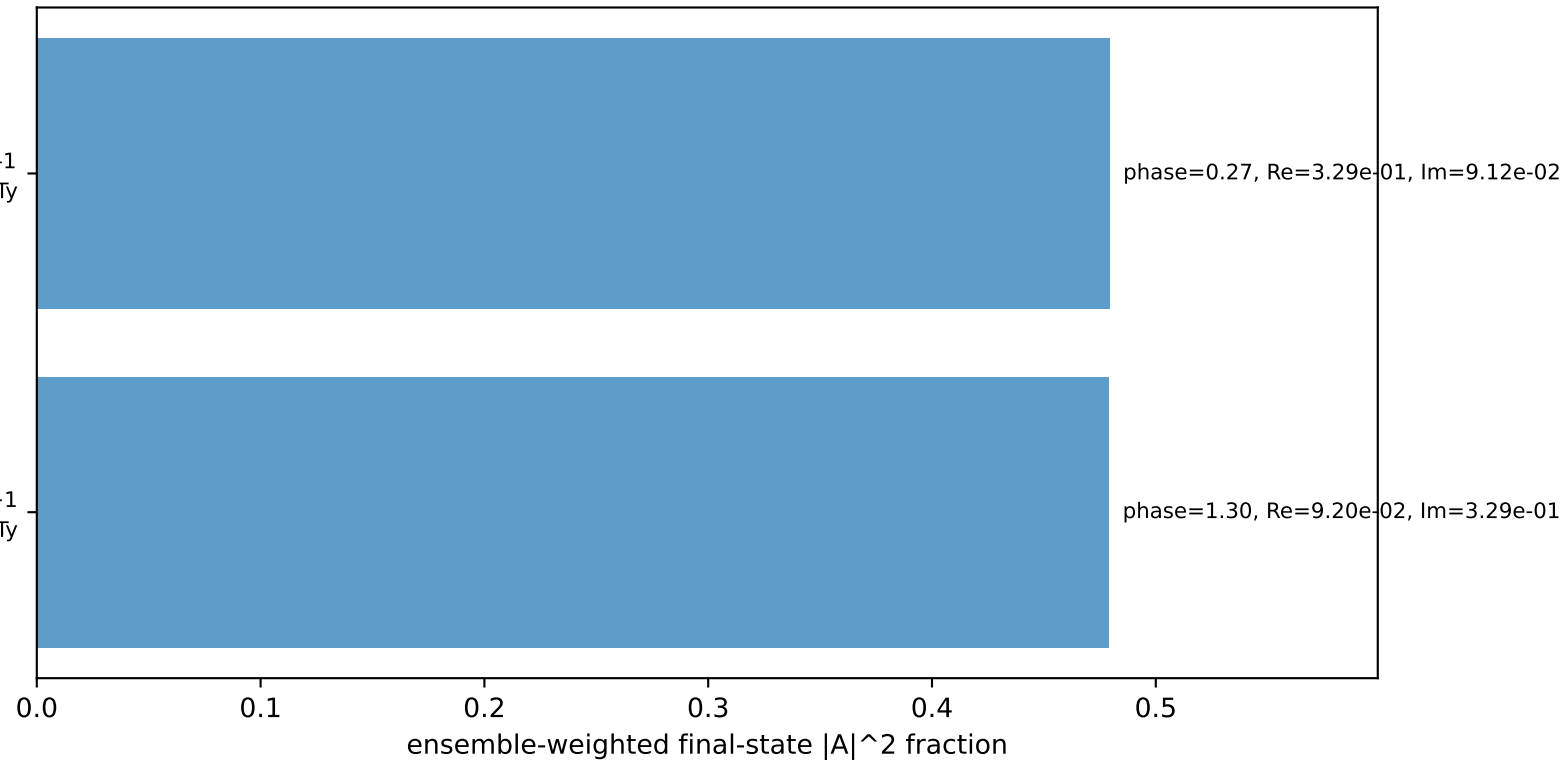
Transverse plane



$h_e=-1, h_p=-1, h_\gamma=-1$
 muon $h=-1$, proton Ty

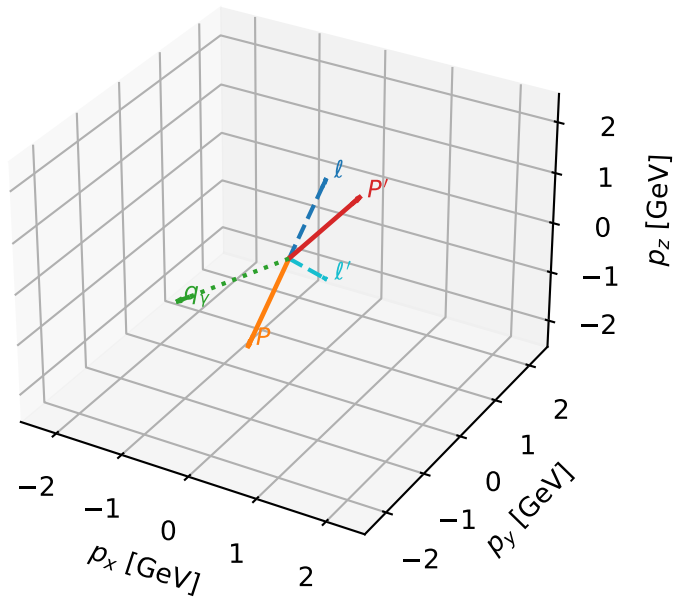
$h_e=+1, h_p=+1, h_\gamma=+1$
 muon $h=+1$, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=95.8%)



Ty proton characteristic max $P_{\text{GHZ+}}$ configuration

3D momenta

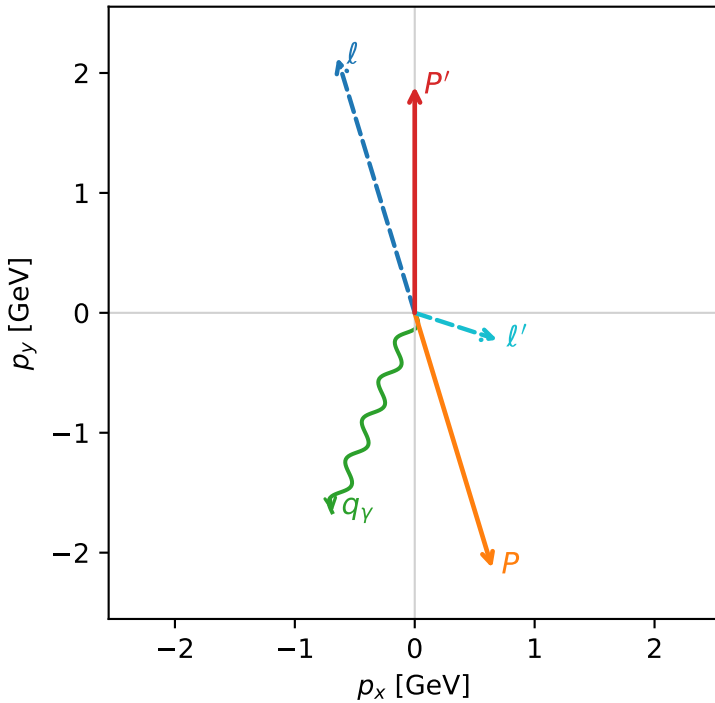


ghz_purity_high_egamma_ty_proton_region_2 $P_{\{\text{rm GHZ+}\}}=0.478826$
 region: high_Egamma_Ty_proton_region_2
 outgoing-state purity: 0.500168
 kinematic point: high_s_high_theta_in_high_Egamma
 incoming state: $\frac{1}{2} \sum_{h_\mu} \rho(h_\mu, T_{y,p})$

$s=21.5158$, $\sqrt{s}=4.63852$, $|\vec{P}|=2.22311$, $|\vec{P}'|=1.88636$
 $\theta_{\text{in}}=1.57079$, $\phi_l=1.86532$, $\phi_p=5.00691$
 $E_\gamma=1.8$, $\phi_\gamma=4.31969$
 $Q^2=5.07463$, $x_B=0.268034$, $t=-16.4329$, $W^2=14.738$, $y=0.917958$
 $\theta(l', \gamma)=94.5328$ deg

four-momenta $[E, p_x, p_y, p_z]$ GeV:
 ℓ $E= 2.2256$ $p_x= -0.6453$ $p_y= 2.1274$ $p_z= -0.0000$
 P $E= 2.4129$ $p_x= 0.6453$ $p_y= -2.1274$ $p_z= 0.0000$
 ℓ' $E= 0.7318$ $p_x= 0.6888$ $p_y= -0.2234$ $p_z= 0.0000$
 P' $E= 2.1067$ $p_x= 0.0000$ $p_y= 1.8864$ $p_z= 0.0000$
 q_γ $E= 1.8000$ $p_x= -0.6888$ $p_y= -1.6630$ $p_z= 0.0000$

Transverse plane



$h_e=-1, h_p=-1, h_\gamma=-1$
 muon $h=-1$, proton Ty

$h_e=+1, h_p=+1, h_\gamma=+1$
 muon $h=+1$, proton Ty

Leading initial-ensemble/final-state components (N=2, retained=95.8%)

